ECON 580: Honors Research Tutorial

Empirical methods

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Plan for today

- Interpreting regression equations: typically, how can we establish causality?
- Tools: difference-in-differences, instrumental variables
  - Lots of quick examples

A Regression Equation

- Take a simple regression of outcome Y_i for unit i on explanatory variable X_i:

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i$$

- The OLS estimator for β_1 is:

$$\widehat{\beta_1} = \widehat{Cov}(X, Y) / \widehat{V}(X)$$

- This is simply a formula. **It does not necessarily provide anything meaningful**
- These statistic are what we call **descriptive statistics** which don't necessarily have any **causal** interpretation

A Regression Equation

How do we interpret regression equations? $Y_i = \beta_0 + \beta_1 X_i + \epsilon_i$

- Without further assumptions, regression equations don't really say much:

$$Y = \text{something observed} + \text{something else not observed}$$

- The crucial issue is how ϵ relates to X
- The first step is to understand what is in ϵ–what is unobserved
- The next step is to understand whether these unobserved factors are **related/correlated with** X (the variables you do observe)
  - If $cov(\epsilon, X) \neq 0$ then you have an endogeneity concern and cannot necessarily claim to estimate a causal relationship

Returns to Schooling Example

- Economists are interested in estimating the labor market "return" to schooling
- Our **economic model** of wages postulates that an individual i's wage Y_i is determined by her years of schooling X_i and ability A_i:

$$Y_i = \beta X_i + \alpha A_i$$

- β and α are parameters that reflect the "return" to schooling and ability, resp.
- To turn this economic model into an **econometric** or **empirical model**, we need to specify **what is observed in data and what is unobserved**
  - We observe Y_i, X_i for each individual (these are **observables**)
  - We do not observe A_i (this is an **unobservable**)

Returns to Schooling Example

- Given A_i is not observed, we can re-write our model of wage determination as:

$$Y_i = \beta X_i + \epsilon_i$$

where $\epsilon_i = \alpha A_i$

- The OLS estimand can then be written as

$$\begin{aligned}\beta(OLS) =& \frac{Cov(Y,X)}{V(X)} = \frac{Cov(\beta X + \epsilon, X)}{V(X)} \\ =& \beta\frac{Cov(X,X)}{V(X)} + \frac{Cov(\epsilon,X)}{V(X)} = \beta + \alpha\frac{Cov(A,X)}{V(X)}\end{aligned}$$

- The OLS estimator is biased for β ($\beta(OLS) \neq \beta$) if $\alpha \neq 0$ (ability produces earnings) **and** $Cov(A,X) \neq 0$ (ability related to schooling)

Ways to Express the Bias

1. Omitted Variable Bias
   - The bias in this case stems from the fact that we cannot observe all aspects of an individual's human capital (schooling and ability)
   - Ability A is therefore **omitted** from the regression model we can estimate
   - If we were able to observe ability completely, then we could include it in our regression model, and the omitted variable bias would be eliminated

Ways to Express the Bias

2. Self-Selection Bias
   - This is also often called **self-selection bias** because the source of the bias is from the fact that individual's **self-select** into schooling, i.e. schooling is not randomly assigned
   - This focuses on the $Cov(A, X)$ part, and if $Cov(A, X) = 0$ (as for example with random assignment of schooling X), then the bias is eliminated and $\beta(OLS) = \beta$ **even if** $\alpha \neq 0$ (ability matters for wages)

Ways to Express the Bias

3. Endogenous Regressors
   - Still another term for this bias is to say that schooling is an **endogenous regressor**
   - That is, schooling is determined **within the model** (by ability and other factors) rather than being determined **exogenously** from outside the model
   - The full model could be

$$Y_i = \beta X_i + \alpha A_i$$

and

$$X_i = \gamma A_i + \eta_i$$

where η_i are other factors that affect schooling

Ways to Express the Bias

4. Correlation is Not Necessarily Causation
   - Finally, we can also think of this bias as reflecting the fact that **observed correlations do not necessarily reflect causation**
   - For almost any data set, the sample covariance between wages and schooling is positive: $Cov(Y, X) > 0$
   - However the self-selection bias suggests that this correlation may not be entirely due to the **causal effect** of schooling on wages given by β

$$Cov(Y, X) = Cov(\beta X + \alpha A, X) = \beta V(X) + \alpha Cov(X, A)$$

Approaches

1. Assume Away the Unobserved Population Heterogeneity
   - The first approach is to assume away the unobserved population heterogeneity by assuming that there is **no unobserved population heterogeneity** in ability: $A_i = A$ for all i
   - With this assumption, we have (adding i subscript to emphasize heterogeneity):

$$Cov(X_i, A_i) = Cov(X_i, A) = 0$$

     (Correlation of random variable X_i with any constant A is 0)
   - The assumption here is that **all individuals have the same level of ability**, and therefore this level of ability would only affect the level of wages, not the slope/return with respect to schooling

Approaches

2. Randomize the Treatment in a Controlled Experiment
   - Imagine if we could assign schooling level X by a coin flip, i.e. conduct a schooling **experiment**
   - Randomization implies $\epsilon \perp X$, $Cov(\epsilon, X) = 0$, and $\beta(OLS) = \beta$
   - Notice that **randomization does NOT imply that each individual has the same level of ability** (as above), but instead that ability is uncorrelated with X
   - This extreme kind of randomization would be difficult in practice–instead one could randomly provide some subsidy for schooling
   - And, although randomization may unquestionably imply **internal validity** (within the experiment sample), **external validity** (extrapolation) to other populations depends on the representativeness of the experiment sample

Approaches

3. Selection on Observables/Matching
   - Another approach identifies the return to schooling by assuming that **selection is only on observable variables**
   - The simplest way to understand this is that if we can measure A exactly, we can then estimate a multivariate regression "controlling" for A (estimate both β and α)
   - Even if we cannot measure A exactly, we might think about conditioning on a set of covariates (e.g. test score values), and arguing that, conditional on these covariates, X is uncorrelated with the residual variation

Approaches

4. IV or Natural Experiments
   - The fourth approach uses **exclusion restrictions**, generating instruments Z
   - These instruments (or often natural experiments) create a source of exogenous variation, like randomized experiments but with additional issues
   - An example of a natural experiment in the schooling case might be a policy which offered some individuals a tuition subsidy to attend school ($Z = 1$ if offered the subsidy, $Z = 0$ if not offered)
   - The **difference-in-difference** method can be seen as an example of an instrument

Approaches

- One common theme across methods is that each method takes a stand on which groups are good comparisons for each other
- For experiments, the randomly assigned treatment and control groups are good comparisons: they are otherwise similar except for some random coin-flip treatment allocation
- Other methods have a similar idea: instruments or natural experiments create comparison groups from naturally occurring variation, matching methods do so using covariates in the dataset, difference-in-difference does so using past experiences, and so on
- This idea of identifying comparison groups is a useful one to understand your particular method

Difference in differences (DD) and instrumental variables (IV)

We are going to focus on two approaches:

1. **Difference-in-differences (DD)**
2. **Instrumental variables (IV)**

Difference-in-differences

Perhaps the most famous diff-in-diff: Card and Krueger (1994) min. wage

New Jersey raised the minimum wage in 1992 from \$4.25 to \$5.05

- What is the effect of raising the minimum wage on fast food employment?

Before vs. After Comparison in NJ:

- Compares: same individuals/communities before and after program
- Drawback: does not control for time trends

Treated state (NJ) vs Control state (PA) Comparisons:

- Compares: employment rates in NJ vs PA after 1992
- Drawback: does not control for permanent employment differences between the two states

Two Wrongs Sometimes Make a Right

Difference-in-differences estimation combines the (flawed) pre vs. post and treated vs control approaches

- This can sometimes overcome the twin problems of [1] selection bias (on fixed traits) and [2] time trends in the outcome of interest
- The basic idea is to observe the treatment group and a comparison group before and after the program

The diff-in-diff estimator is:

$$DD = \overline{Y}_{post}^{treatment} - \overline{Y}_{pre}^{treatment} - \left(\overline{Y}_{post}^{comparison} - \overline{Y}_{pre}^{comparison}\right)$$

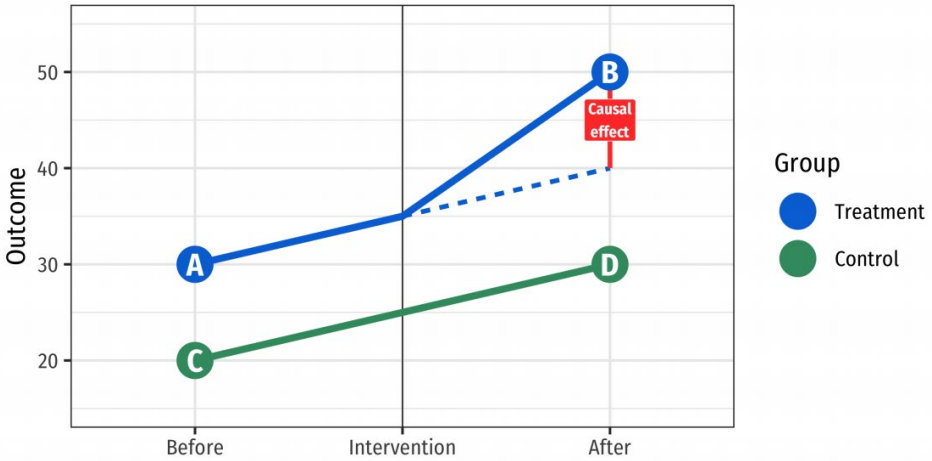
In a regression framework this would look like:

$$Y_{i,t} = \alpha + \beta Treat_i + \gamma Post_t + \delta(Treat_i \times Post_t) + \varepsilon_{i,t}$$

	Pre mean	Post mean	
Control	A (never treated)	B (never treated)	
Treatment	C (not yet treated)	D (treated)	

	Pre mean	Post mean	Δ (post – pre)
Pennsylvania	23.33 A	21.17 B	-2.16 B – A
New Jersey	20.44 C	21.03 D	0.59 D – C
Δ (NJ – PA)	-2.89 C – A	-0.14 D – B	(0.59) – (–2.16) = 2.76

Graphical depiction of a diff-in-diff



Diff-in-diff in a regression framework

$$Y_{it} = \alpha + \beta\,\text{Group}_i + \gamma\,\text{Time}_t + \delta\,(\text{Group}_i \times \text{Time}_t) + \varepsilon_{it}$$

α = Mean of control, pre-treatment

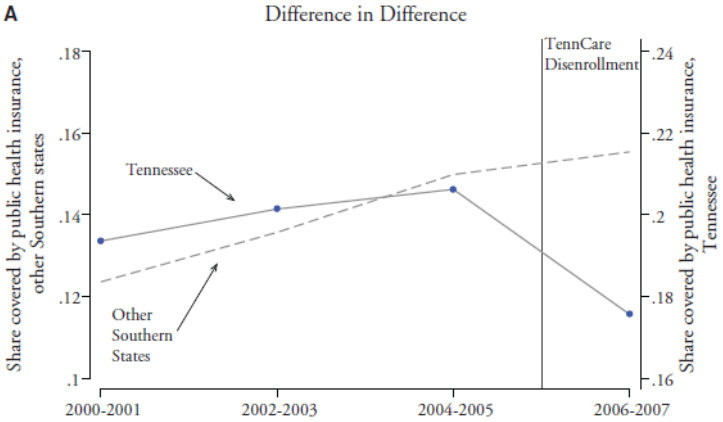
β = Increase in outcome across groups

γ = Increase in outcome over time within units

δ = Difference in differences!

	Pre mean	Post mean	Δ (post – pre)
Control	α	α + γ	γ
Treatment	α + β	α + β + γ + δ	γ + δ
Δ (trtmt – ctrl)	β	β + δ	δ

Diff-in-diff example: TennCare disenrollment



Diff-in-Diff in a Panel Data Framework

- We can also implement diff-in-diff when we have more than two periods of data; this can increases statistical power

$$Y_{i,t} = \alpha_i + \gamma_t + \beta Treat_{i,t} + \varepsilon_{i,t}$$

- We can also estimate **event studies** which include dummies for the effect of treatment in each time period

$$Y_{i,t} = \alpha_i + \gamma_t + \beta_t Treat_{i,t} + \varepsilon_{i,t}$$

- Moreover, sometimes treatment occurs at different times for different people, locations, ... i.e., variation in treatment timing. Framework above allows for this
- Lots of examples of state level policies occur at different times in different states
- We can think of this variation similarly to a bunch of diff-in-diffs stacked together

Variation in treatment timing: Food Stamp rollout

Example: counties introduced food stamps at different times

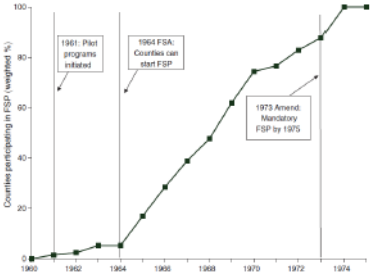


FIGURE 1. WEIGHTED PERCENT OF COUNTIES WITH FOOD STAMP PROGRAM, 1960–1975

Source: Authors' tabulations of food stamp administrative data (US Department of Agriculture, various years). Counties are weighted by their 1960 population.

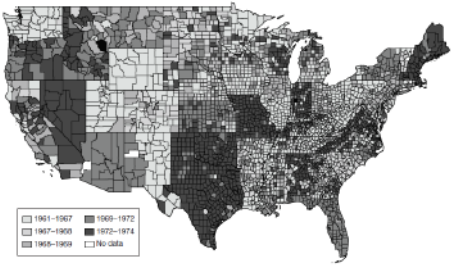


FIGURE 2. FOOD STAMP PROGRAM START DATE, BY COUNTY, 1961–1974

Notes: Authors' tabulations of food stamp administrative data (US Department of Agriculture, various years). The shading corresponds to the county FSP start date, where darker shading indicates later county implementation.

Source: Almond, Hoynes, and Schanzenbach (AER, 2016)

Variation in treatment timing: Medicaid rollout

Example: states adopted Medicaid at different times

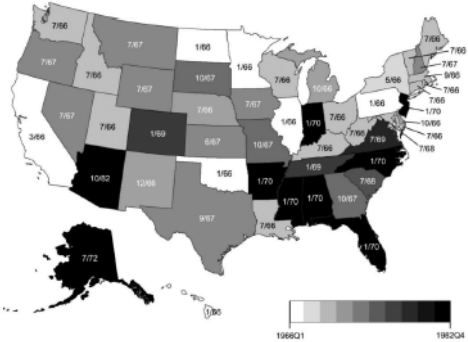


Figure 2.
Medicaid Adoption by Quarter
Notes: Adoption dates come from the Department of Health Education and Welfare (1970) & Social Security Administration (2013). The map is shaded relative to the quarter of adoption and states are labeled with the month and year of adoption.

Source: Boudreaux, Golberstein, and McAlpine (Journal of Health Economics, 2016)

Variation in treatment timing: Community Health Centers rollout

Example: counties opened community health centers at different times

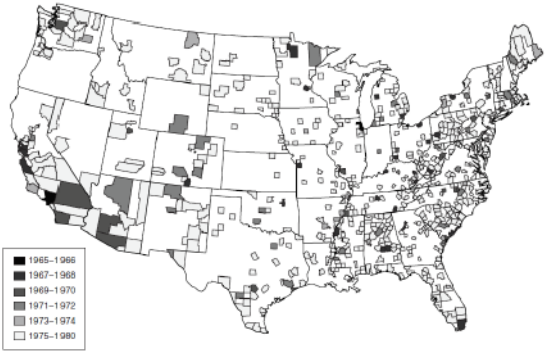
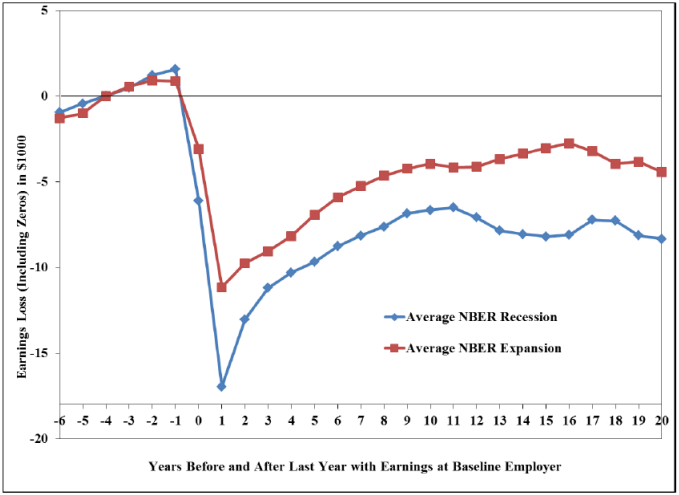


FIGURE 3. ESTABLISHMENT OF COMMUNITY HEALTH CENTERS BY COUNTY OF SERVICE DELIVERY, 1965–1980

Note: Dates are the first year that a CHC was established in the county.
Source: Information on CHCs drawn from NACAP and PHS reports.

Source: Bailey and Goodman-Bacon (AER, 2015)

Event study: cost of job loss



Source: Davis and von Wachter (2011, BPEA).

Instrumental Variables

Does education cause higher earnings?

$$\text{Earnings}_i = \beta_0 + \beta_1 \text{Education}_i + \epsilon_i$$

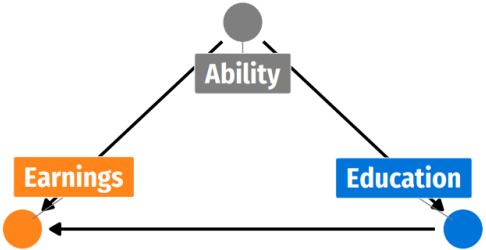
Outcome variable

Policy/program variable

If we ran this regression, would beta1 give us the causal effect of education?

Does education cause higher earnings?

- It would likely **not** give us the causal effect of education on earnings
- Why? Education is correlated with many other variables that affect earnings
  - In other words, education is "endogenous"
  - For example, ability affects both education and earnings, so the regression of education on earnings is also picking up the effect of ability on earnings



Does education cause higher earnings?

Endogeneity

The error term (ϵ) is related to the explanatory variables

$$\text{Earnings}_i = \beta_0 + \beta_1 \text{Education}_i + \epsilon_i$$

Education is related to some part of this this unobserved stuff ϵ

Education is partly endogenous (because it is picking up ability)

Instruments: fixing endogeneity problems

- How do we filter out this endogenous part of education to estimate just the exogenous effect of education on earnings?
- We could control for the endogeneity by adding ability to our regression:

$$\text{Earnings}_i = \beta_0 + \beta_1 \text{Education}_i + \beta_2 \text{Ability} + \epsilon_i$$

- But ability is often unobserved!
- Instead, we typically rely on an *instrument*

- Instrument: something that is correlated with the policy variable (education), but does not directly affect the outcome (except through the policy variable)

Instrumental variables: terminology and example

- **First stage / relevance:** Instrument Z affects education
- **Exclusion restriction:** Z *only* affects earnings indirectly through effect on educ

Example (Angrist and Krueger 1991):

- Most countries have compulsory schooling laws, e.g., can't drop out until turn 16
- Then **month of birth** is an instrument for education:

Instrumental variables: terminology and example

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Example (Angrist and Krueger 1991):

- Most countries have compulsory schooling laws, e.g., can't drop out until turn 16
- Then **month of birth** is an instrument for education:
  - Children born in Q4: start school just before you turn 6. At age 16 you have completed 10+ years of school
  - Children born in Q1: start school in September after you turn 6. At age 16 you have only completed 9 year + a few months of school
- Month of birth probably doesn't correlate with ability (exogeneity!)

Instrumental variables: example 2

- What is the effect of going to war on future earnings? (Angrist Krueger 1994)
- Problem with just regressing earnings on veteran status:

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- In other words, veteran status is likely correlated with unobservables
- Instrument:

Instrumental variables: example 2

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- Problem with just regressing earnings on veteran status: individuals with bad labor market prospects are more likely to join the army and go to war and, independently, have lower future earnings
- In other words, veteran status is likely correlated with unobservables
- Instrument: the draft!
  - Being drafted affects the probability of going to war (relevance)
  - Being drafted is purely random (exogenous)

Instrumental variables: example 3

- What is the effect of having children on female labor force participation?
- Problem with regressing LFP on number of children:

Instrumental variables: example 3

- What is the effect of having children on female labor force participation?
- Problem with regressing LFP on number of children: people who like to have lots of kids just happen to also not have good labor market prospects
- So the number of children is an endogenous regressor
- Instrument ideas:
  - Having twins at second birth (exogenous increase in number of kids)
  - First two children are same sex

Instrumental variables: example 4

- What is the effect of population growth on the number of new firms? (Karahan, Pugsley and Sahin (2023, AER))
- Hard to answer using national data
- Way to get more data: exploit state-level variation in population growth and new firms!
- Problem: states that have higher population growth may have better economies with more new firms ....
- Instrument ideas:
  - Use lagged fertility (20+ years ago) as an instrument for population growth today

Examples of undergraduate research theses

It could be useful to peruse some undergraduate theses (these are from other schools):

- https://www.econ.berkeley.edu/undergrad/theses
- https://economics.stanford.edu/undergraduate/honors-program/honors-theses
- https://econ.duke.edu/undergraduate/current-students/honors-program/showcases-awards
- https://www.econ.umd.edu/undergraduate/selected-honors-theses-recent-graduates